The impact of punctuated infectiousness on infectious disease dynamics

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Abstract

1 Introduction

Epidemic trajectories are profoundly shaped by individual-level effects. For example, interpersonal variation in pathogen shedding and contact rates can lead to superspreading, which impacts both the likelihood that an epidemic will occur and how explosive it will be if it does [Lloyd-Smith et al., 2005]. Likewise, chance events early in an epidemic can modulate when widespread transmission becomes established, durably shifting the outbreak's overall timing [Morris et al., 2024]. Accounting for such effects is vital for public health preparedness: while understanding an epidemic's expected trajectory is useful — and can be done by largely overlooking individual-level effects — assessing variation around that mean, including reasonable best- and worst-case scenarios, requires careful attention to individual-level transmission dynamics.

The mean-field dynamics of an epidemic are determined by two fundamental quantities: the basic reproduction number, R_0, and the intrinsic generation interval distribution, $g(\tau)$ [Wallinga and Lipsitch, 2007], where τ represents time since infection. These can be combined into a single object, the "infectiousness profile" $A(\tau) = R_0\, g(\tau)$, which captures both the expected amount of transmission arising from an index case and its timing, and is the key parameter in the renewal equation framework for epidemic modeling [Breda et al., 2012]. Analogously, at the individual level, the individual reproduction number, ν_i, captures the expected number of infections person i would produce in an otherwise susceptible population [Lloyd-Smith et al., 2005], and the individual-level intrinsic generation interval distribution, $g_i(\tau)$, captures the timing of those infections. These can be combined into the "individual infectiousness profile" or the "infection kernel", $a_i(\tau) = \nu_i g_i(\tau)$ [Champredon and Dushoff, 2015, Svensson, 2007]. The population-level quantities are the expectations of the individual-level ones, with $R_0 = E_i[\nu_i]$, $g(\tau) = E_i[g_i(\tau)]$, and $A(\tau) = E_i[a_i(\tau)]$. Substantial work has gone into understanding how variation in ν_i (*i.e.*, the area under $a_i(\tau)$) impacts epidemic dynamics, but the impact of variation in $g_i(\tau)$ (*i.e.*, how $a_i(\tau)$ is distributed over time), holding R_0 and $g(\tau)$ fixed and without confounding from variation in ν_i, is less well understood.

Since $A(\tau)$ is an expectation over individuals, it need not resemble any single person's $a_i(\tau)$. The standard susceptible-exposed-infectious-recovered (SEIR) model provides a useful illustration: it assumes each person undergoes a latent period of length $x_i \sim \text{Exp}(\gamma_{\text{SEIR}})$ during which they are not infectious, followed by an infectious period of length $y_i \sim \text{Exp}(\alpha_{\text{SEIR}})$ during which they transmit at rate β_{SEIR}. The resulting "stepwise" individual infectiousness profile is

$$a_i(\tau) = \begin{cases} \beta_{\text{SEIR}} & \tau \in [x_i, x_i + y_i] \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Averaging over individuals yields the population-level infectiousness profile

$$A_{\text{SEIR}}(\tau) = E_i[a_i(\tau)] = \beta_{\text{SEIR}} \frac{\gamma_{\text{SEIR}}}{\gamma_{\text{SEIR}} - \alpha_{\text{SEIR}}} (e^{-\alpha_{\text{SEIR}}\tau} - e^{-\gamma_{\text{SEIR}}\tau}) \tag{2}$$

i.e., a smooth, bell-shaped curve that emerges from a collection of discontinuous $a_i(\tau)$ (**Figure XX**). Other choices of $a_i(\tau)$ can also produce the same $A_{\text{SEIR}}(\tau)$: for example, setting $a_i(\tau) \equiv A_{\text{SEIR}}(\tau)$ for all individuals gives a "smooth" profile in which each person's infectiousness follows the population average (**Figure XX**). At the opposite extreme, $a_i(\tau) = (\beta_{\text{SEIR}}/\alpha_{\text{SEIR}})\delta_{\tau_i^*}(\tau)$, with $\delta_{\tau_i^*}(\tau)$ a delta function that has mass at τ_i^* drawn from $g(\tau) = A_{\text{SEIR}}(\tau)/R_0$, gives a "spike" profile in which each person's infectiousness is concentrated at a single moment (**Figure XX**). All three families — stepwise, smooth, and spike — produce the same $A_{\mathrm{SEIR}}(\tau)$ and hence the same deterministic, mean-field epidemic dynamics, but the resulting stochastic dynamics may differ.

While estimates of R_0 and $g(\tau)$, and thus $A(\tau)$, are commonplace for most pathogens, we often have little understanding of $a_i(\tau)$'s shape. Viral kinetics offer an indirect window into the individual-level timing and intensity of infectiousness, but the precise mapping is poorly understood. Contact tracing data are often used to inform ν_i, but capturing $g_i(\tau)$ from the same data is challenging due to uncertainty in infection onset times and imperfect case ascertainment. Nevertheless, we can reason about the potential shapes of $a_i(\tau)$: the full variability in $g(\tau)$ must arise from both within-individual and between-individual variation in infection timing, suggesting that $a_i(\tau)$ is generally narrower than $A(\tau)$. Some evidence indicates that, for SARS-CoV-2 and influenza, the window of peak transmissibility at the individual level may be extremely short — even less than a day — which is substantially narrower than the pathogens' overall generation interval distributions Goyal et al. [2021]. So, while estimates of $a_i(\tau)$ are lacking, the available evidence suggests that the shape of $a_i(\tau)$ may differ meaningfully from that of $A(\tau)$.

However, a central question remains: to what extent does the shape of $a_i(\tau)$, holding $A(\tau)$ fixed, impact epidemic dynamics? Several lines of evidence suggest it could matter substantially, but the question has not been studied systematically. Within the SEIR framework (which assumes a stepwise individual infectiousness profile), replacing exponentially-distributed latent and infectious periods with more realistic distributions can dramatically alter epidemic trajectories, changing peak size, epidemic duration, growth rate, outbreak probability, and estimates of R_0, even when the mean periods are held fixed [Lloyd, 2001, Wearing et al., 2005, Britton and Lindenstrand, 2009]. Replacing the stepwise $a_i(\tau)$ with a time-varying, viral kinetics-informed infectiousness profile can generate significant variability in epidemic peak size and timing [Li et al., 2023]. In applied settings, the shape of $a_i(\tau)$ enters implicitly: models of contact tracing effectiveness depend on "the course of individual infectivity" as a key input [Klinkenberg et al., 2006], and models of testing-based interventions rely on assumptions about the connection between viral load and infectiousness, which generally produce individual infectiousness profiles that are substantially narrower than the population average [Larremore et al., 2021]. Yet in each of these cases, the shape of $a_i(\tau)$ is either a modeling assumption adopted for a different purpose or a feature varied at the population level; no study has systematically isolated the impact of variation in $g_i(\tau)$ on epidemic dynamics while holding R_0, $g(\tau)$, and $A(\tau)$ constant at the population level.

Here, we address this gap by conducting a systematic analysis of how the shape of $a_i(\tau)$ (*i.e.*, the decomposition of $g(\tau)$'s variability into within- *vs.* between-individual components) affects epidemic behavior, holding all population-level quantities constant. To do so, we introduce a flexible modeling framework, parameterized by a smoothness parameter $\psi \in [0, 1]$, that interpolates between the "spike" and "smooth" extremes. This approach extends the renewal equation framework to accommodate individual-level variation in infectiousness timing [Breda et al., 2012]. We assess how the shape of $a_i(\tau)$, independent of other factors, creates variation in epidemic establishment times, impacts the inference of the epidemic growth rate r and the generation interval distribution $g(\tau)$, and introduces additional overdispersion in secondary infection counts. We also determine how $a_i(\tau)$'s shape modulates the effectiveness of detect-and-isolate interventions and gathering size restrictions. We illustrate these findings using three respiratory pathogens — influenza, SARS-CoV-2 omicron, and measles — but emphasize that the methods and findings are general. We also address the question of whether the shape of $a_i(\tau)$ can be inferred from contact-tracing data, and conclude with a discussion of implications for epidemiological modeling and outbreak control.

2 Modeling framework

2.1 *Accounting for within- vs. between-individual variation in infectiousness timing: the time-shifted Gamma model*

We begin by introducing a modeling framework that decomposes $g(\tau)$'s variance into within- and between-individual components *via* a single parameter. We seek an analytically tractable definition of $g_i(\tau)$ such that $g(\tau) = E_i[g_i(\tau)]$, and where the width of $g_i(\tau)$ is governed by a smoothness parameter $\psi \in [0,1]$ that interpolates between the "spike" and "smooth" extremes.

To achieve this, we first assume that $g(\tau)$ follows a Gamma(α, β) distribution with rate $\beta = \alpha/\bar{g}$, where $\bar{g}$ is the mean generation time. The Gamma distribution is a natural and common choice for modeling generation interval distributions. Next, we define each person's individual intrinsic generation interval distribution as

$$g_i(\tau) = \begin{cases} 0 & \tau \leq l_i \\ f_\epsilon(\tau - l_i) & \tau > l_i \end{cases} \tag{3}$$

where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ is a random latent period and f_ϵ is the Gamma$(\psi\alpha, \beta)$ density (**Figure XX**). Equivalently, in terms of random variables, the secondary infections j from an index case i are distributed at times $\tau_{ij} = l_i + \epsilon_j$, where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ and $\epsilon_j \overset{iid}{\sim} \text{Gamma}(\psi\alpha, \beta)$.

Since l_i and ϵ_j are independent Gamma-distributed random variables with common rate β, their sum follows a $\text{Gamma}((1-\psi)\alpha + \psi\alpha, \beta) = \text{Gamma}(\alpha, \beta)$ distribution; thus $E_i[g_i(\tau)] = g(\tau)$, as required. Furthermore, when $\psi \to 1$, the latent period vanishes ($l_i \to 0$) and f_ϵ approaches the full Gamma(α, β) density, which gives the "smooth" extreme (no between-individual variation). Similarly, when $\psi \to 0$, f_ϵ concentrates to a point mass and $g_i(\tau)$ becomes a delta function at $l_i \sim \text{Gamma}(\alpha, \beta)$, which gives the "spike" extreme (no within-individual variation).

2.2 *Accounting for time-varying contacts*

For $A(\tau)$ and $a_i(\tau)$ to be functions of τ (time since infection) alone, and not also of calendar time t, it is necessary to assume that interpersonal contact rates do not vary over time. We follow this convention for most of our analyses. However, the interaction between punctuated $g_i(\tau)$ and time-varying contacts may be consequential. Intuitively, if a person's infectiousness is concentrated in a narrow window, then their realized transmission depends heavily on their contact level during that window; whereas if their infectiousness is spread over a long period, contact fluctuations will get averaged out. Thus, we briefly introduce a generalization that accounts for time-varying contacts. We also introduce two specific contact models that serve as test beds.

Under time-uniform contacts, $a_i(\tau) = \nu_i g_i(\tau)$, where ν_i and $g_i(\tau)$ are defined with respect to a reference contact level which we denote $\bar{c} \equiv 1$. To allow for time-varying contacts, we introduce a contact process $\mathbf{c}(t)$: a nonnegative, wide-sense stationary stochastic process with time-mean $E_t[\mathbf{c}(t)] = 1$. Each person i has a contact modulator $c_i(t)$ that is a realization of $c(t)$, representing the proportional scaling of that person's contact rate relative to the reference level at calendar time t. The constraint $E_t[\mathbf{c}(t)] = 1$ is an ensemble average over all possible realizations at time t; individual realizations c_i need not have time-average 1, allowing for individuals with persistently higher or lower contact rates than the population average. We additionally define the "biological infectiousness", b_i, as the expected number of secondary infections person i would produce under $\bar{c} \equiv 1$. The individual infectiousness profile thus generalizes to

$$a_i(t_i, \tau) = b_i g_i(\tau) c_i(t_i + \tau) \tag{4}$$

where t_i is person i's infection time. If $c(t) \equiv \bar{c} = 1$, this reduces back to the familiar $a_i(\tau) = \nu_i g_i(\tau)$, with $b_i = \nu_i$. More generally,

$$\nu_i = b_i \int_0^\infty g_i(\tau) c_i(t_i + \tau)\, d\tau \tag{5}$$

i.e., the individual reproduction number is a function of the person's inherent biological infectiousness b_i as well as the interaction between their infectiousness timing and their contact process. Likewise, the realized generation interval distribution is

$$g_i^{\text{re}}(t_i, \tau) = \frac{g_i(\tau) c_i(t_i + \tau)}{\int_0^\infty g_i(u) c_i(t_i + u)\, du} \tag{6}$$

So, an equivalent rendering of Eq. 4 is

$$a_i(t_i, \tau) = \nu_i g_i^{\text{re}}(t_i, \tau) \tag{7}$$

For the population-level quantities, it can be shown that

$$R_0 = E_i[\nu_i] = E_i[b_i] \qquad \text{and} \qquad g(\tau) = E_i[b_i g_i(\tau)]/R_0 \tag{8}$$

where the second equality follows from the assumed independence of (b_i, g_i) and c_i and the constraint $E[\mathbf{c}(t)] = 1$ (**Supplementary Methods**). This gives the population-level infectiousness profile

$$A(t, \tau) = R_0 g(\tau) C(t + \tau) \tag{9}$$

where $C(t) = E_i[c_i(t)]$, the population-average contact level at calendar time t.

As test cases, we consider two models. We define the "periodic" contact model

$$c_i(t) = 1 - c_{amp} \cos\left(\frac{2\pi t}{c_{per}}\right) \quad \forall i \tag{10}$$

with amplitude $c_{amp} \in [0, 1]$ and period $c_{per} > 0$ (**Figure XX**). In this model, all individuals share the same deterministic contact function; this is a degenerate case of the contact process $\mathbf{c}(t)$ in which every realization is identical. While there is temporal variation within each person's contacts, there is no variation across individuals. The process' stationarity is achieved by considering a random infection time t_i which is uniformly distributed over one period (which may approximately hold during the period of exponential epidemic growth) [Papoulis, 1965]. At the population level, $C(t) = E_i[c_i(t)] = 1 - c_{amp} \cos(2\pi t / c_{per})$.

We also define the "stochastic" contact model, where each person's $c_i(t)$ is a piecewise-constant process that switches to a new level at arrivals of a Poisson process with rate λ (**Figure XX**). The contact levels are drawn independently from a Gamma(σ, σ) distribution, which has mean 1 (satisfying the constraint $E_t[\mathbf{c}(t)] = 1$) and variance $1/\sigma$. This creates a two-parameter family indexed by the dispersion parameter $\sigma > 0$ and switching rate λ. Smaller σ yields more heterogeneous contacts, while large σ yields a bell-shaped distribution concentrated around 1. As $\sigma \to \infty$, the distribution collapses to a point mass at 1, recovering the constant-contact reference process. In contrast to the periodic model, each person's contact trajectory is unique, but the population average is constant in time: $C(t) = E_i[c_i(t)] = 1$.

2.3 *Simulation approach*

To investigate the impact of individual-level infectiousness timing on epidemic dynamics, we use both closed-form analytic results and stochastic epidemic simulations. We initialized simulations with a single infectious individual in a finite population of $N = 10,000$. For each simulation, we drew a latent period l_i from a Gamma$((1-\psi)\alpha, \beta)$ distribution and, where relevant, a contact trajectory $c_i(t)$ for each person i. Upon infection, the number of secondary infection attempts χ_i was drawn from a Poisson(b_i) distribution. Throughout, we assume $b_i \equiv R_0$, *i.e.*, no interpersonal variation in biological infectiousness, to isolate the effect of infectiousness timing from that of variation in the individual reproduction number. For each infection attempt j, a random time offset ϵ_j was drawn from a Gamma$(\psi\alpha, \beta)$ distribution, so that the timing of the attempt was $\tau_{ij} = l_i + \epsilon_j$. To account for time-varying contacts, we applied Poisson thinning, accepting each attempt with probability proportional to $c_i(t_i + \tau)$. Each attempted infection was then paired with a uniformly chosen individual from the population: if the recipient was susceptible, infection occurred; otherwise, the attempt failed.

Full simulation details are provided in the **Supplementary Methods**. For illustration, we focused on three sets of epidemiological parameters reflecting published estimates of R_0 and $g(\tau)$ of influenza, SARS-CoV-2 omicron, and measles (**Table XX**).

3 The impact of punctuated infectiousness on uncontrolled epidemics

3.1 *Punctuated infectiousness profiles generate later and more variably-timed epidemics.*

Early-outbreak stochasticity can impact when an epidemic becomes established, which in turn shifts the epidemic's full trajectory [Morris et al., 2024]. Simulations demonstrate that narrower $a_i(\tau)$ (*i.e.*, $\psi \to 0$) cause the epidemic establishment time to become more variable and to shift modestly later in expectation. For 5,000 simulations across each of the three benchmark pathogens and for $\psi \in \{0, 0.5, 1\}$, we measured the time to reach a cumulative prevalence of 5% (500 total infections), which we used as a marker of epidemic establishment. For influenza, the mean [5% quantile, 95% quantile] time to reach 500 infections was 24.2 days [17.8, 33.8] for $\psi = 1$ *vs.* 24.5 days [17.2, 34.9] for $\psi = 0$; for omicron, 18.1 days [14.8, 22.9] for $\psi = 1$ *vs.* 20.3 days [13.4, 30.4] for $\psi = 0$; and for measles, 31.3 days [28.6, 34.6] for $\psi = 1$ *vs.* 32.5 days [26.4, 39.5] for $\psi = 0$. This translated into meaningful differences in epidemic timing, particularly for SARS-CoV-2 omicron and measles: on day 20, 82% of simulated SARS-CoV-2 omicron outbreaks with $\psi = 1$ had reached 500 cases, while only 55% of outbreaks with $\psi = 0$ had reached the same threshold (**Figure XX**). Similarly, by day 33, 84% of simulated measles outbreaks with $\psi = 1$ had reached 500 cases, while only 58% of outbreaks with $\psi = 0$ had reached the same threshold (**Figure XX**).

These trends can be understood intuitively by considering the two extremes of ψ. When infectiousness is maximally punctuated ($\psi = 0$), all of an index case's secondary infections occur at a single moment $\tau_{ij} l_i$, a single draw from $g(\tau)$. When infectiousness is maximally smooth ($\psi = 1$), the secondary infection times are independent draws from $g(\tau)$. For the smooth profile, the earliest secondary infection is the minimum order statistic of χ_i independent draws from $g(\tau)$, which falls earlier in expectation than a single draw. This order statistic effect is magnified as R_0, and thus the typical χ_i, increases. Since the speed at which an epidemic establishes depends disproportionately on the earliest transmissions in each generation, smooth profiles confer an effective head start. Conversely, under the punctuated profile, the entire next generation's timing is determined by a single random quantity l_i; thus, an unusually early or late draw propagates fully into the epidemic trajectory without being offset by more "typical" draws, and inflates the variability in epidemic establishment times.

Analytic results confirm these trends. Following Morris et al. [2024], we describe the cumulative number of infections $Z(t)$ at time t during the early-epidemic exponential growth phase as a rescaled branching process

$$W(t) = e^{-rt} Z(t) \tag{11}$$

with initial condition $W(0)$ and growth rate r. The process W converges almost surely to a random variable W as $t \to \infty$. W can be loosely interpreted as a random initial condition on the branching process that captures the effect of early-outbreak stochasticity: large values of W correspond to random successful early transmission events, inflating the effective initial condition. With a change of variables, this random initial condition can be expressed as a random time shift ς, where

$$\varsigma = \frac{\log W - \log E[W]}{r} \tag{12}$$

and the branching process can be expressed as

$$Z(t) \approx e^{r(t+\varsigma)} \tag{13}$$

For notational simplicity, we assume that epidemics are always seeded with a single infectious case; thus $E[W] = Z_0 = 1$ and $\varsigma = \log W/r$.

The characterization of W and its transformation into a time shift ς follow standard theory [Morris et al., 2024]. Building upon this foundation, we can derive $\text{Var}[W]$ as an explicit function of ψ (**Supplementary Methods**). When $g(\tau)$ follows the Gamma time-shift model, the variance of the random initial condition is:

$$\text{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^{\alpha} + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} := \frac{\sigma_1^{\alpha} + \sigma_2^{(1-\psi)\alpha} - 1}{1 - \sigma_1^{\alpha}} \tag{14}$$

where $z = R_0^{1/\alpha} - 1$ and $\sigma_c = ((1+z)^c)/(1+2z)$. Thus, $\text{Var}[W]$ depends on R_0, α (the shape/skewness of $g(\tau)$), and ψ. The punctuation parameter ψ enters Eq. 14 only once, in the exponent of σ_2. Since $\sigma_2 > 1$ when $R_0 > 1$, $\text{Var}[W]$ is monotonically decreasing in ψ, so that narrower individual infectiousness profiles ($\psi \to 0$) maximize the variance. Furthermore, σ_2 is monotonically increasing in R_0 (**Supplementary Methods**), so larger R_0 amplifies the sensitivity of $\text{Var}[W]$ to ψ.

If we assume, following Morris et al. [2024], that the distribution of $W|\text{surv}$ (*i.e.*, conditioning on epidemic non-extinction) is well-approximated by a moment-matched Gamma distribution, we can translate Eq. 14 into implications for the time shift ς. Specifically, since ς is the logarithm of an approximately Gamma-distributed random variable, we can express $E[\varsigma|\text{surv}]$ and $\text{Var}[\varsigma|\text{surv}]$ in terms of the digamma and trigamma functions (**Supplementary Methods**). These expressions confirm that narrower infectiousness profiles, which inflate $\text{Var}[W]$, also lead to more negative $E[\varsigma|\text{surv}]$ (*i.e.*, cause the expected time shift to fall later) and higher $\text{Var}[\varsigma|\text{surv}]$.

These findings also do not require us to assume that $g(\tau)$ follows the Gamma time-shift model. For any probability densities f_l and f_ϵ that define an individual latent period and infectiousness kernel, such that $f_l(\tau) * f_\epsilon(\tau) = g(\tau)$ where $*$ denotes convolution, it can be shown that shifting variance from f_ϵ to f_l increases $\text{Var}[W]$, with the effect maximized by the spike profile and modulated by R_0 (**Supplementary**

Methods). If we assume the distribution of W is well-approximated by a Gamma distribution, the effects of f_ϵ's variance on $E[\varsigma|\text{surv}]$ and $\text{Var}[\varsigma|\text{surv}]$ follow as before.

3.2 Punctuated infectiousness profiles yield less certain estimates of the epidemic growth rate and generation interval distribution.

During an emerging epidemic, inferring the growth rate and the generation interval distribution are among the first critical tasks. These are key ingredients for calculating the reproduction number, which determines the effort needed to control an outbreak. Simulations indicate that punctuated individual infectiousness profiles cause estimates of r to be less certain. For each of the three benchmark pathogens, we simulated 5,000 epidemics in infinite populations. We ran each simulation for 7 days after reaching 100 cumulative cases and binned the infections into daily counts. Then, to estimate the growth rate, we fit a Poisson generalized linear model using maximum likelihood to the daily case counts. For all three pathogens, punctuated infectiousness profiles yielded more uncertain growth rate estimtes, with the ψ-related uncertainty increasing in R_0 (**Figure XX**). For influenza, the maximum likelihood growth rate estimate had standard deviation XX for $\psi = 1$, XX for $\psi = 0.5$, and XX for $\psi = 0$; for SARS-CoV-2 omicron, the growth rate estimate had standard deviation XX for $\psi = 1$, XX for $\psi = 0.5$, and XX for $\psi = 0$; and for measles, the growth rate estimate had standard deviation XX for $\psi = 1$, XX for $\psi = 0.5$, and XX for $\psi = 0$.

This uncertainty in r follows from overdispersion in the daily case counts generated by more punctuated individual infectiousness profiles (**Figure XX**). To see why, consider person i, infected at time t_i, who generates $\chi_i \sim \text{Poisson}(R_0)$ total secondary infections. Let $N_i(\tau)$ be the number of these infections that fall on day-since-infection $[\tau, \tau+1)$, and let $Q_i(\tau)$ be the probability that a given secondary infection from person i lands on that day:

$$Q_i(\tau) = \int_\tau^{\tau+1} f_\epsilon(u - l_i)\, du \tag{15}$$

where l_i is person i's realized latent period. The expectation of $Q_i(\tau)$ over latent period draws is

$$E[Q_i(\tau)] = \int_\tau^{\tau+1} g(u)\, du \equiv q(\tau) \tag{16}$$

which depends only on the population-level generation interval distribution and is therefore independent of ψ. Conditional on $Q_i(\tau)$, the Poisson thinning property gives $N_i(\tau) \mid Q_i(\tau) \sim \text{Poisson}(R_0\, Q_i(\tau))$. Applying the law of total variance over $Q_i(\tau)$ yields the index of dispersion (**Supplementary Methods**):

$$\text{ID}[N_i(\tau)] = \frac{\text{Var}[N_i(\tau)]}{E[N_i(\tau)]} = 1 + R_0 \frac{\text{Var}[Q_i(\tau)]}{E[Q_i(\tau)]} \tag{17}$$

The entire ψ-dependence enters through $\text{Var}[Q_i(\tau)]$. In the smooth case ($\psi = 1$), $f_\epsilon = g$ and $l_i = 0$ for all individuals, so $Q_i(\tau) = q(\tau)$ is deterministic: $\text{Var}[Q_i(\tau)] = 0$ and $\text{ID} = 1$. In the spike case ($\psi = 0$), f_ϵ is a point mass, so $Q_i(\tau)$ is Bernoulli, equaling 1 with probability $q(\tau)$ and 0 otherwise: $\text{Var}[Q_i(\tau)] = q(\tau)(1 - q(\tau))$ and $\text{ID} = 1 + R_0(1 - q(\tau))$. More generally, $\text{Var}[Q_i(\tau)]$ is monotonically decreasing in ψ (**Supplementary Methods**), so narrower infectiousness profiles always yield more overdispersed daily counts. Since infectors contribute independently, this overdispersion carries through to the aggregate daily infection count, inflating its variance by a factor that scales with R_0 (**Supplementary Methods**). The standard Poisson GLM assumes equidispersed counts; when the data are in fact overdispersed, the sampling variability of $\hat{r}$ across epidemic realizations exceeds the nominal Poisson-based precision, consistent with the simulation results above.

Simulations also indicate that punctuated individual infectiousness profiles cause estimates of $g(\tau)$ to be less certain. Typically, $g(\tau)$ is estimated using contact tracing data capturing serial intervals, from which generation intervals are inferred by assuming an observation delay model. The generation intervals are typically assumed to represent *iid* draws from the generation interval distribution, which becomes increasingly inaccurate for narrower individual infectiousness profiles. In the limit of $\psi \to 0$, all generation intervals from a given index case are identical. The effective sample size thus ranges from roughly KR_0 when $\psi = 1$, where K is the number of index cases, to K when $\psi = 0$. Observation delays can obscure the correlation between secondary infection times, making it easy to assume that variation in secondary infection times from a single infector are due to differences in infection timing, when in fact they are only due to differences in the observation model.

Simulations demonstrate the impact of this phenomenon. We simulated offspring from 5,000 index cases for each of $\psi \in \{0, 0.2, 0.5, 0.7, 1\}$ and for each of the three benchmark pathogens. We applied a Gamma(a_{obs}, b_{obs}) observation error independently to each infection time (both index and secondary cases) to generate a set of simulated serial intervals. We then use these serial intervals to calculate the estimated generation interval distribution parameters, assuming that the serial intervals were independent. This represents a sort of statistically "best possible" scenario: we assumed ascertainment was perfect (all secondary cases from a given index cases were identified, and it was known which case was the index case and which were the secondary cases, even if a serial interval was negative); and we assumed perfect knowledge of the delay distribution parameters. Even so, we identified meaningful differences in the estimates of the generation interval distribution parameters (α, β) across ψ-values.

[[results]]

3.3 *Punctuated infectiousness profiles interact with time-varying contact patterns to generate overdispersion in secondary infection counts*

Superspreading arises when secondary infection counts are overdispersed, *i.e.*, when the variance in the number of secondary infections produced by an individual exceeds the mean. This overdispersion can powerfully impact epidemic dynamics, making epidemics less likely to establish but potentially more explosive when they do [Lloyd-Smith et al., 2005].

Mathematically, overdispersion in secondary infection counts is captured through variation in the individual reproduction number, ν_i. Secondary infection counts are typically assumed to follow a Poisson distribution: $\chi_i \sim \text{Poisson}(\nu_i)$. When $\nu_i \equiv R_0$ (no variation across individuals), $\text{Var}[\chi_i] = E[\chi_i] = R_0$, *i.e.*, there is no overdispersion. When ν_i itself is variable, this inflates the variance of χ_i relative to the mean, creating overdispersion. In the overdispersed scenario, secondary infection counts are often modeled using a Negative Binomial distribution:

$$\chi_i \sim \text{NegBin}(R_0, k) \tag{18}$$

with mean $E[\chi_i] = R_0$ and variance $\text{Var}[\chi_i] = R_0(1 + R_0/k)$. Thus, $k \in (0, \infty)$ captures overdispersion, with $k \to \infty$ yielding Poisson-distributed χ_i and $k \to 0$ yielding greater overdispersion. For convenience, we introduce the parameter $\text{OD} = 1/k$, so that large OD corresponds to high overdispersion.

From Eq. 5, we see that variation in ν_i can arise in three fundamental ways: variation in biological infectiousness, b_i; variation in individual contact levels, $E_t[c_i(t)]$; and the variable interaction between infectiousness timing and the contact function, $g_i(\tau)c_i(t_i + \tau)$. The first two contributors to overdispersion (so-called "super-shedders" and "super-contactors") have been studied extensively; here, we examine the impact of the $g_i - c_i$ interaction as a third mechanism of superspreading, holding $b_i \equiv R_0$ constant and $E_t[c_i(t)] = 1$.

Simulations demonstrate that narrower $g_i(\tau)$ yield more overdispersion in the presence of time-varying contacts. For the periodic contact function with a period of $c_{per} = 1$ day and amplitude $c_{amp} = 0.5$, empirical estimates of k ranged from XX at $\psi = 1$ to XX at $\psi = 0$ for influenza; from XX at $\psi = 1$ to XX at $\psi = 0$ for SARS-CoV-2 omicron; and from XX at $\psi = 1$ to XX at $\psi = 0$ for measles. For the stochastic contact process, when $k_c = 10$ (corresponding to xx), k ranged from XX at $\psi = 1$ to XX at $\psi = 0$ for influenza; from XX at $\psi = 1$ to XX at $\psi = 0$ for SARS-CoV-2 omicron; and from XX at $\psi = 1$ to XX at $\psi = 0$ for measles. This led to meaningful differences in early-outbreak extinction probabilities: for influenza, the extinction probability was XX for $\psi = 0$ *vs.* XX for $\psi = 1$; for SARS-CoV-2 omicron, the extinction probability was XX for $\psi = 0$ *vs.* XX for $\psi = 1$; and for measles, the extinction probability was XX for $\psi = 0$ *vs.* XX for $\psi = 1$.

For the periodic contact model, using results from signal processing theory, we can derive a simple closed-form expression for $\text{OD} = 1/k$ in terms of the model parameters (**Supplementary Methods**):

$$\text{OD} = \frac{c_{amp}^2}{2}\left(1 + \left(\frac{2\pi\bar{g}}{\alpha c_{per}}\right)^2\right)^{-\psi\alpha} \tag{19}$$

where c_{amp} and c_{per} are the amplitude and period of the contact process, respectively. This reveals a few key features about the relationship between ψ and k. First, as expected, $c_{amp} \to 0$ yields $\text{OD} \to 0$, *i.e.*, in the degenerate case where contacts do not vary over time, we recover the standard Poisson-distributed secondary infection distribution with no overdispersion. Also, OD is strictly decreasing in ψ, *i.e.*, narrower infectiousness profiles ($\psi \to 0$) yield more overdispersion. The sensitivity of OD to ψ, however, depends on the magnitude of $\eta = 2\pi\bar{g}/(\alpha c_{per})$. When $\eta \gg 1$, which happens for example when the contact process' period is much shorter than the mean generation time ($c_{per} \ll \bar{g}$), OD is small regardless of ψ. In this case, the contact process varies so quickly that virtually all infectiousness profiles average over the entire contact process, yielding little overdispersion. Alternatively, when $\eta \ll 1$, which can happen when $c_{per} \gg \bar{g}$, $\text{OD} \approx c_{amp}^2/2$ regardless of ψ. In this case, the contact process varies so slowly that all infectiousness profiles, whether narrow or broad, "see" only a small part of the contact process' variation, causing maximal overdispersion that is insensitive to the shape of the individual infectiousness profile.

The impact of ψ is therefore most visible when $\eta \approx 1$, in which case $\psi \to 1$ causes the infectiousness profile to effectively average over the contact variation, while $\psi \to 0$ translates the contact variation into meaningful overdispersion. For our benchmark pathogens, $\bar{g}$ ranges from 3.2 days to 12.2 days. Thus, weekly variation in contacts ($c_{per} = 7$) causes OD to vary meaningfully with ψ. Daily variation ($c_{per} = 1$) gets averaged out, yielding little overdispersion unless $\psi \approx 0$. Monthly variation ($c_{per} = 30$) leads to $\text{OD} \approx c_{amp}^2/2$ regardless of ψ. While the analytic form is less revealing, similar patterns hold for the stochastic contact model, with the switching rate λ and dispersion parameter σ playing analogous roles to c_{per} and c_{amp}.

4 The impact of punctuated infectiousness profiles on controlled epidemics.

4.1 Detect-and-isolate interventions.

Detect-and-isolate interventions are among the most important non-pharmaceutical interventions available for curbing an epidemic. The goal of these interventions is to detect individuals at an early stage of infection, prior to peak infectiousness, so that they can shift their behavior (isolate) and prevent onward spread. Detection can happen *via* the development of symptoms and/or diagnostic testing; thus, the timing of symptom onset and the window of test-based detectability, relative to the infectiousness profile, are critical factors influencing the success of such interventions [Fraser].

The efficacy of detect-and-isolate interventions can be summarized in terms of *testing effectiveness* (TE), defined as the expected proportion by which a detect-and-isolate intervention reduces population-level transmission [Middleton and Larremore, 2024]. Formally, TE can be expressed as

$$\mathrm{TE} = \rho\eta P(\tau > D) \tag{20}$$

where $\rho \in [0,1]$ is the participation rate, $\eta \in [0,1]$ is the effectiveness of isolation (so that post-detection transmission attempts succeed with probability $1-\eta$), τ is the time of a transmission attempt relative to the index case's infection time, and D is the detection time. When D is anchored to the index case's infection time (*e.g.*, the detectability window spans 2–7 days after infection), TE depends only on the shape of the generation interval distribution; the shape of the individual infectiousness profile integrates out (**Supplementary Methods**). If, however, D is anchored to the individual infectiousness profile (*e.g.*, the window of detectability extends $\pm$ 3 days from peak infectiousness), TE depends on the width of the individual infectiousness profile, as we illustrate here. Anchoring D to the infectiousness profile is arguably the more realistic scenario since symptom onset, test-based detectability, and peak pathogen shedding are all linked to viral load, which in turn shapes the individual's infectiousness trajectory [Boyer].

To illustrate how TE depends on the shape of the individual infectiousness profile, we considered two detection mechanisms: symptom-based isolation and regular screening with a diagnostic test. For the symptom-based detection mechanism, we assumed symptoms arose at a Normally-distributed time relative to peak infectiousness with mean offset δ_{symp} and standard deviation σ_{symp}, with isolation occurring immediately upon developing symptoms. For the testing-based strategy, we assumed the window of detectability ran from δ_{pre} days before peak infectiousness to δ_{post} days after peak infectiousness, and that diagnostic tests were administered at even intervals of length Δ_{test} with uniform-random offset relative to the start of the detectability window. We included a $\mathrm{Exp}(\lambda_{act})$-distributed test turnaround delay before isolation. To cover various widths of the infectiousness profile, we considered $\psi \in \{0, 0.2, 0.5, 0.8, 1\}$. For each scenario, we numerically integrated the expression $\mathrm{TE} = \eta\rho \int_{-\infty}^{\infty} (1 - F_\epsilon(t)) f_{\text{det}}(t)\, dt$, where F_ϵ is the CDF of the infectiousness kernel ($f_\epsilon \sim \mathrm{Gamma}(\psi\alpha, \beta)$) and f_{det} is the PDF of the detection time distribution, with time indexed from the peak of the individual infectiousness profile.

These experiments demonstrate that narrow infectiousness profiles ($\psi \to 0$) make TE sensitive to small shifts in detection time. When symptoms arise at a $\mathrm{Normal}(\delta_{\text{symp}}, \sigma^2_{\text{symp}})$-distributed time after peak infectiousness, the drop in TE as δ_{symp} shifts from negative to positive (average detection happens before *vs.* after peak infectiousness) is sharpest for narrow infectiousness profiles (**Figure XX**). When $\sigma_{\text{symp}} = 0.5$ and $\delta_{\text{symp}} = 3$, TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, XX for $\psi = 1$ to XX for $\psi = 0$ against measles. When $\delta_{\text{symp}} = -1$, the trends reversed: TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, XX for $\psi = 1$ to XX for $\psi = 0$ against measles. Similarly, when the window of test-based detectability extended from $\delta_{\text{pre}} = 3$ days prior to peak infectiousness to $\delta_{post} = 7$ days after, and when testing occurred weekly with an $\text{Exp}(0.5)$-distributed turnaround time, TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, and XX for $\psi = 1$ to XX for $\psi = 0$ against measles. When testing occurred daily, TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, and XX for $\psi = 1$ to XX for $\psi = 0$ against measles. Incomplete participation and imperfect isolation scale these effects, but the qualitative patterns remain (**Supplementary Figure XX**). In short, these findings reveal how narrower profiles cause detect-and-isolate protocols to become increasingly "all-or-nothing", where a detection prior to peak infectiousness averts a person's entire infectious potential while a detection after peak infectiousness averts none.

Beyond the direct reduction in transmission captured by TE, detect-and-isolate interventions produce two additional ψ-dependent effects that act in opposite directions: generation interval truncation (which can accelerate epidemic growth) and increased overdispersion in secondary infection counts (which increases the probability of epidemic extinction). First, detect-and-isolate interventions re-shape the generation interval distribution and cause the mean generation time to contract — an effect that is most pronounced for smooth profiles and disappears as $\psi \to 0$. Under a detect-and-isolate protocol, the generation interval distribution becomes

$$\text{[Distorted GI]} g^*(\tau) = \tag{21}$$

For smooth infectiousness profiles, detection reliably cuts off the tail of transmission; any surviving transmission attempts thus fall earlier in expectation, shortening the mean generation interval. For punctuated profiles, the all-or-nothing nature of detection causes some individuals' infectiousness to be cut off entirely, while others are fully missed — and, among those that are missed, their secondary infection times follow $g^*(\tau) = g(\tau)$ on average, yielding no shortening of the mean generation interval. Since shorter mean generation times with fixed R_{eff} yield faster epidemic growth [Wallinga and Lipsitch, 2007], epidemics controlled by detect-and-isolate interventions are expected to grow faster than those simulated "naively" with reproduction number $R_0(1 - TE)$ and generation interval $g(\tau)$, with the smoothest individual infectiousness profiles yielding the greatest discrepancies.

Second, detect-and-isolate interventions also increase overdispersion in secondary infection counts, with punctuated infectiousness profiles yielding the most pronounced increases in overdispersion. In the $\psi \to 0$ limit, the variance of the individual reproduction number ν_i is maximized, with

$$\nu_i = \begin{cases} 0 & \text{w.p.TE} \\ R_0 & \text{w.p.} 1 - \text{TE} \end{cases} \tag{22}$$

i.e., a distribution consisting of two point masses at 0 and R_0. Overdispersion increases the probability of early-epidemic extinction, so epidemics controlled by detect-and-isolate interventions are expected to go extinct more frequently than those simulated naively with matching reproduction number $R_0(1 - TE)$ and generation interval $g(\tau)$, with the most punctuated individual infectiousness profiles yielding the highest extinction probability.

Simulations confirm these expected trends. We simulated epidemics under detect-and-isolate interventions, assuming 50% participation ($\rho = 0.5$), perfect isolation ($\eta = 1$), and a deterministic isolation time

1 day before peak infectiousness (equivalent to $\delta_{\text{symp}} = -1$ and $\sigma_{\text{symp}} = 0$). We also simulated an analogous set of epidemics generated by setting $R_{\text{eff}} = (1 - \text{TE})R_0$, but keeping the generation interval $g(\tau)$ unchanged and introducing no overdispersion ($\nu_i \equiv R_0$). We ran 5,000 epidemic simulations in a population of size 10,000 for each of $\psi \in \{0, 0.5, 1\}$ and for each of the three benchmark pathogens. For the detect-and-isolate simulations, the mean ([5% quantile, 95% quantile]) time to reach 500 infections was XX [XX, XX] for influenza, XX [XX, XX] for SARS-CoV-2 omicron, and XX [XX, XX] for measles; while for the naive TE-adjusted simulations the time to reach 500 infections was XX [XX, XX] for influenza, XX [XX, XX] for SARS-CoV-2 omicron, and XX [XX, XX] for measles. The proportion of epidemics that never reached 500 cases in the detect-and-isolate simulations was XX for influenza, XX for SARS-CoV-2 omicron, and XX for measles; and in the naive TE-adjusted simulations, the proportions were XX for influenza, XX for SARS-CoV-2 omicron, and XX for measles.

Together, these findings reveal that the impact of a detect-and-isolate intervention on an outbreak's trajectory depends sensitively, and through multiple mechanisms, on the shape of the individual infectiousness profile. For smooth individual infectiousness profiles, the reduction in R_{eff} captured by the testing effectiveness can be partially counteracted by generation interval truncation, which can cause the epidemic to grow faster than one might expect. For punctuated individual infectiousness profiles, the reduction in R_{eff} may be compounded by elevated overdispersion in secondary case counts, so that epidemics are not only less intense, but also less likely to establish in the first place.

4.2 Gathering size restrictions.

Gathering size restrictions are another important class of non-pharmaceutical interventions, aimed at limiting opportunities for major transmission events. Gathering size restrictions truncate the "tail" of the interpersonal contact distribution, reducing the mean contact rate (and thus the effective reproduction number) as well as reducing overdispersion. Since the individual infectiousness profile's shape is connected with overdispersion in the presence of time-varying contacts, there is reason to believe that gathering size restrictions may play out differently for punctuated *vs.* smooth individual infectiousness profiles. Here, we show that gathering size restrictions yield two main effects: a reduction in mean transmission that is independent of the infectiousness profile's shape, and a reduction in overdispersion whose magnitude depends on ψ.

To investigate this, we considered a modification of the stochastic contact process. As before, the contact level $c_i(t)$ for person i switches at $\text{Exp}(\lambda)$-distributed times, but the new levels are drawn from a truncated $\text{Gamma}(\sigma, \sigma)$ distribution, rather than from the full distribution. The truncation threshold, $c_{\max}$, captures the severity of the restriction.

Truncation reduces the effective reproduction number by the same factor regardless of the individual infectiousness profile's shape. The effective reproduction number is

$$R_{\text{eff}} = \mu_T R_0 \qquad \text{where} \qquad \mu_T = \frac{F_{\text{Gamma}}(c_{\max}; \sigma + 1, \sigma)}{F_{\text{Gamma}}(c_{\max}; \sigma, \sigma)} \tag{23}$$

where $F_{\text{Gamma}}(c; a, b)$ is the $\text{Gamma}(a, b)$ CDF evaluated at c.

It is possible to derive simple expressions for the relative and absolute difference in overdispersion between the uncontrolled and the gathering-size-restricted scenarios (**Supplementary Methods**). These are

$$\frac{\mathrm{OD}_{\mathrm{new}}}{\mathrm{OD}_{\mathrm{old}}} = \frac{\sigma}{\mu_T^2/V_T} \qquad \text{and} \qquad \mathrm{OD}_{\mathrm{new}} - \mathrm{OD}_{\mathrm{old}} = \mathrm{OD}_{\mathrm{old}}\left[\frac{\sigma}{\mu_T^2/V_T} - 1\right] \tag{24}$$

where V_T is the variance of the Gamma(σ, σ) distribution truncated at c_{max}. The relative difference in overdispersion does not depend on the individual infectiousness profile's shape; however, the absolute difference does, insofar as $\mathrm{OD}_{\mathrm{old}}$ depended on the punctuation parameter ψ. Concretely: if $\mathrm{OD}_{\mathrm{old}}$ is small (≈ 0), which is more likely for smooth individual infectiousness profiles, then $\mathrm{OD}_{\mathrm{new}} - \mathrm{OD}_{\mathrm{old}} \approx 0$, *i.e.*, gathering size restrictions cause little practical difference in an already-small overdispersion. If, however, $\mathrm{OD}_{\mathrm{old}}$ is large, which is more likely for punctuated individual infectiousness profiles, the absolute difference in overdispersion may be meaningful: gathering size restrictions may cause a previously large $\mathrm{OD}_{\mathrm{old}}$ to shrink to a meaningfully smaller $\mathrm{OD}_{\mathrm{new}}$.

To investigate the impact of gathering size restrictions on outbreak trajectories, we simulated 5,000 epidemics in a population of size 10,000 using the stochastic contact model with $\sigma = XX$ and with gathering sizes truncated at $c_{\mathrm{max}} = XX$. We also simulated two parallel sets of "naive" epidemic simulations, both with $R_{\mathrm{eff}} = \mu_T R_0$: one with the full (non-truncated) stochastic contact model, and one with time-uniform contacts ($c_i(t) \equiv 1$). We considered $\psi \in \{0, 0.5, 1\}$ for the three benchmark pathogens. Across simulations, the R_{eff}-adjusted, stochastic contact model over-estimated the epidemic extinction probability and the R_{eff}-adjusted, time-uniform contact model under-estimated it relative to the truncated contact model, and the discrepancies were largest for the narrowest individual infectiousness profiles ($\psi \to 0$) (**Figure XX**). In the $\psi = 0$ boundary case, the proportion of simulated influenza epidemics that failed to establish was XX for the truncated contact model, XX for the R_{eff}-adjusted full stochastic contact model, and XX for the R_{eff}-adjusted time-uniform contact model; for SARS-CoV-2 omicron, XX for the truncated contact model, XX for the R_{eff}-adjusted full stochastic contact model, and XX for the R_{eff}-adjusted time-uniform contact model; and for measles, XX for the truncated contact model, XX for the R_{eff}-adjusted full stochastic contact model, and XX for the R_{eff}-adjusted time-uniform contact model.

Similar to detect-and-isolate interventions, these findings indicate that gathering size restrictions can have additional overdispersion-mediated impacts on epidemic trajectories that go beyond their reduction of mean contact rates, and these impacts are sensitive to the shape of the individual infectiousness profile. Previously, we showed that punctuated profiles generate more overdispersion under time-varying contacts; here, we find that gathering size restrictions attenuate this overdispersion, with the largest absolute reductions occurring precisely when the baseline overdispersion is highest, *i.e.*, with narrow individual infectiousness profiles. Failing to account for this can lead to either over- or under- estimates of the epidemic extinction probability, depending on which model is used.

5 Identifiability and inference of the individual infectiousness profile's shape

Focus here on R_0 and uncertainty in infection timing – that should be enough.

6 Discussion

- For the ς section, the shift in $E[\varsigma]$ is minor; the inflation of Var$[\varsigma]$ is the real story.
- For the sensitivity of TE: note that shifts in symptom timing are commonplace (happend with SARS-CoV-2), and small changes in testing frequency are easy operational decisions.

- Flesh out the counter-intuitive impact of TE/GE contraction/overdispersion for detect-and-isolate, and Reff/overdispersion for gathering size restrictions.

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7 Supplementary Materials

7.1 *Supplementary Methods*

7.1.1 Deriving population-level quantities from individual-level components

7.1.2 Simulation approach

We used stochastic, individual-based epidemic simulations to investigate the impact of individual-level infectiousness timing on epidemic dynamics. The simulation model has a modular design: an infection attempt generator produces a set of transmission attempts for each infected individual, and a population-level simulation script processes these attempts to propagate the epidemic. We implemented two simulation variants: a finite-population model with susceptible depletion, and an infinite-population model where every infection attempt succeeds. Both are implemented in **R** (version 4.5.0), with performance-critical inner loops reimplemented in C++ via **Rcpp** for speed.

Infection attempt generation. For each newly infected individual i (with infection time t_i), the infection attempt generator proceeds as follows:

1. **Draw the number of infection attempts.** The number of secondary infection attempts χ_i is drawn from a Poisson(b_i) distribution, where b_i is the individual's biological infectiousness. Throughout, we set $b_i \equiv R_0$ for all individuals, so that variation in realized transmission arises solely from infectiousness timing and contact dynamics.

2. **Draw the latent period.** A latent period l_i is drawn from a Gamma$((1-\psi)\alpha, \beta)$ distribution, representing the delay between infection and the onset of transmissibility. When $\psi = 1$, the latent period is zero; when $\psi = 0$, it accounts for the full variance of the generation interval.

3. **Draw the timing of each infection attempt.** For each attempt $j = 1, \ldots, \chi_i$, an independent jitter ϵ_j is drawn from a Gamma$(\psi\alpha, \beta)$ distribution. The time of infection attempt j relative to the index case's infection time is $\tau_{ij} = l_i + \epsilon_j$. When $\psi = 0$, $\epsilon_j = 0$ and all attempts occur simultaneously at time l_i; when $\psi = 1$, the ϵ_j are independent draws from the full Gamma(α, β) generation interval distribution.

4. **Thin for time-varying contacts (if applicable).** When a time-varying contact process $c_i(t)$ is active, each infection attempt at calendar time $t_i + \tau_{ij}$ is accepted with probability $c_i(t_i + \tau_{ij})/c_{\max}$, where $c_{\max}$ is the supremum of the contact process, using Poisson thinning. For the periodic contact model (Eq. 10), all individuals share the same deterministic contact function $c_i(t) = 1 - c_{\text{amp}} \cos(2\pi t/c_{\text{per}})$, and $c_{\max} = 1 + c_{\text{amp}}$. For the stochastic contact model, each individual's piecewise-constant contact trajectory is generated by drawing switching times from a Poisson process with rate λ and contact levels from a Gamma(σ, σ) distribution; the contact trajectory is generated over an interval $[0, 5\bar{g}]$ to ensure enough contact levels are drawn to accommodate all infection attempts. Thinning is performed against the realized maximum contact level over that individual's trajectory.

5. **Thin for detect-and-isolate interventions (if applicable).** When a detect-and-isolate intervention is active, an individual-specific detection time D_i is drawn (relative to peak infectiousness l_i) according to the detection mechanism in effect (symptom-based or screening-based). The individual participates

with probability ρ. If participating, all infection attempts occurring after detection ($\tau_{ij} > t_i + D_i$) are removed with probability η, representing the effectiveness of post-detection isolation.

Finite-population simulation. The finite-population simulation tracks susceptible depletion in a population of size N. It proceeds as follows:

1. **Initialization.** A single index case is infected at time $t = 0$. All other $N-1$ individuals are susceptible.
2. **Event scheduling.** Each infection attempt is stored as a pending event with its scheduled calendar time $t_i + \tau_{ij}$. Events are maintained in a priority queue, ensuring that the earliest pending event is always processed next.
3. **Event processing.** At each step, the earliest event is removed from the queue. A target individual is selected uniformly at random from the population ($N-1$ individuals excluding the infector). If the target is susceptible, infection occurs: the target is marked as infected, and its own infection attempts are generated and queued. If the target is already infected or recovered, the attempt fails ("wasted" on a non-susceptible).
4. **Termination.** The simulation terminates when the event queue is empty (epidemic extinction) or when a stopping criterion is met (*e.g.*, a specified number of infections or calendar time is reached).

Infinite-population simulation. The infinite-population simulation is a special case in which every infection attempt succeeds (no susceptible depletion). This is equivalent to a branching process approximation of the early epidemic and is useful for studying early-epidemic dynamics without the confounding effect of herd immunity. The implementation is identical to the finite-population version except that step 3 is simplified: every queued event produces a new infection, with no target selection or susceptibility check.

Default parameters. Unless otherwise stated, simulations used a population size of $N = 10{,}000$, with $n_{\text{sim}} = 5{,}000$ replicate simulations per scenario. Epidemic establishment was defined as reaching $0.05N = 500$ cumulative infections. To cover a range of infectiousness profile shapes, we considered $\psi \in \{0, 0.2, 0.5, 0.8, 1\}$.

Benchmark pathogens. For illustration, we focused on three sets of epidemiological parameters reflecting published estimates of R_0 and $g(\tau)$ for influenza, SARS-CoV-2 (omicron), and measles. The generation interval distribution $g(\tau) \sim \text{Gamma}(\alpha, \beta)$ was parameterized using the mean generation time $\bar{g}$ and variance $\text{Var}[g]$ reported in the literature, with $\alpha = \bar{g}^2/\text{Var}[g]$ and $\beta = \bar{g}/\text{Var}[g]$.

Pathogen	R_0	$\bar{g}$ (days)	$\text{Var}[g]$ (days^2)	α	β
Influenza	2	3.2	4.41	2.32	0.73
SARS-CoV-2 (omicron)	6	7.05	20.8	2.39	0.34
Measles	12	12.2	13.10	11.36	0.93

Table 1: Benchmark pathogen parameters used in simulations.

7.1.3 *Overdispersion in daily infection counts from punctuated infectiousness profiles*

Here we derive the index of dispersion of the daily infection count attributable to a single infector, show that it is monotonically decreasing in the punctuation parameter ψ, and show that the overdispersion aggregates to the population level.

Setup. Fix an index case i, infected at time t_i, who generates $\chi_i \sim \text{Poisson}(R_0)$ total secondary infections. Under the Gamma time-shift model, the generation interval for each secondary infection is $\tau = l_i + \epsilon$, where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ is a shared latent period and $\epsilon \sim \text{Gamma}(\psi\alpha, \beta)$ is an independent individual jitter. Let $N_i(\tau)$ be the number of person i's secondary infections that fall on day-since-infection $[\tau, \tau+1)$, and define the random landing probability

$$Q_i(\tau) = \int_\tau^{\tau+1} f_\epsilon(u - l_i)\, du \tag{25}$$

where f_ϵ is the Gamma$(\psi\alpha, \beta)$ density. The randomness in $Q_i(\tau)$ arises from the draw of l_i.

Mean of $Q_i(\tau)$. Taking the expectation over l_i and applying Fubini's theorem:

$$E[Q_i(\tau)] = \int_\tau^{\tau+1} E_{l_i}[f_\epsilon(u - l_i)]\, du = \int_\tau^{\tau+1} (f_l * f_\epsilon)(u)\, du = \int_\tau^{\tau+1} g(u)\, du \equiv q(\tau) \tag{26}$$

since $f_l * f_\epsilon = g$ by construction. Thus $E[Q_i(\tau)]$ depends only on the population-level generation interval distribution and is independent of ψ.

Index of dispersion of $N_i(\tau)$. Conditional on $Q_i(\tau)$, the χ_i offspring are independently assigned to day $[\tau, \tau+1)$ with probability $Q_i(\tau)$. By the Poisson thinning property (marginalizing over χ_i):

$$N_i(\tau) \mid Q_i(\tau) \sim \text{Poisson}(R_0\, Q_i(\tau)) \tag{27}$$

Applying the law of total variance:

$$\begin{aligned} \text{Var}[N_i(\tau)] &= E[\text{Var}[N_i(\tau) \mid Q_i(\tau)]] + \text{Var}[E[N_i(\tau) \mid Q_i(\tau)]] \\ &= R_0\, E[Q_i(\tau)] + R_0^2\, \text{Var}[Q_i(\tau)] \end{aligned} \tag{28}$$

Since $E[N_i(\tau)] = R_0\, E[Q_i(\tau)] = R_0\, q(\tau)$, the index of dispersion is

$$\text{ID}[N_i(\tau)] = \frac{\text{Var}[N_i(\tau)]}{E[N_i(\tau)]} = 1 + R_0 \frac{\text{Var}[Q_i(\tau)]}{E[Q_i(\tau)]} \tag{29}$$

Extreme cases. In the smooth case ($\psi = 1$), $f_\epsilon = g$ and $l_i = 0$ deterministically, so $Q_i(\tau) = q(\tau)$ is constant across individuals. Thus $\text{Var}[Q_i(\tau)] = 0$ and $\text{ID}[N_i(\tau)] = 1$.

In the spike case ($\psi = 0$), f_ϵ is a point mass at zero. Then $Q_i(\tau) = \mathbf{1}(l_i \in [\tau, \tau+1))$, which is Bernoulli with parameter $q(\tau)$. Thus $\text{Var}[Q_i(\tau)] = q(\tau)(1 - q(\tau))$ and

$$\text{ID}[N_i(\tau)] = 1 + R_0(1 - q(\tau)) \tag{30}$$

7.2 Supplementary Figures